Huawei Access Node YEL-ACC-17 Optical Transport Failure Alarm Resolution SOP

Node: YEL-ACC-17 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_8240

Description

The YEL-ACC-17 Access Node is a critical component of the Yelahanka Sub-station network infrastructure, responsible for providing high-speed optical transport services to connected end-users. This node utilizes Huawei's advanced access network equipment, which employs a sophisticated combination of SFP (Small Form-factor Pluggable) and XFP (10Gbps Small Form-factor Pluggable) interfaces for optical signal transmission and reception. The SFP/XFP interfaces are responsible for transmitting and receiving optical signals across the fiber optic cables, which are susceptible to degradation and power loss due to environmental factors such as temperature fluctuations, vibration, and aging. In the event of an Optical Transport Failure (OTF) on the YEL-ACC-17 Access Node, the ALARM_8240 alarm is triggered, indicating a critical failure in the optical transport domain. This SOP outlines the steps necessary to diagnose and resolve the root cause of the OTF, ensuring minimal downtime and optimal network performance.

Pre-requisite Checks

- Check: Verify that the Yelahanka Sub-station is accessible and that remote hands are dispatched to the site., Description: Ensure that the site is secure and that all necessary safety precautions are taken before proceeding with the diagnostic and resolution steps.

- Check: Confirm the status of the redundant link on the YEL-ACC-17 Access Node., Description: Verify that the redundant link is operational and ready to take over in the event of a failure.

- Check: Verify the power supply to the YEL-ACC-17 Access Node., Description: Confirm that the power supply is stable and within the acceptable range.

Potential Root Causes

Cause	Probability
SFP/XFP Interface Degradation	High
Fiber Cut or Damage	Medium
Rx/Tx Power Loss due to Environmental Factors	Low

Diagnostic Commands & Expected Outputs

```
Command: display optics interface brief, Expected Output: Optics Interface Brief Information, Description: This command displays a brief summary of the optics interface status, including the SFP/XFP interface type, link status, and receive/transmit power levels.
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Command: display optics interface detailed, Expected Output: Optics Interface Detailed Information, Description: This command displays detailed information about the optics interface, including the SFP/XFP interface type, link status, receive/transmit power levels, and error counters.

Command: display optics interface error, Expected Output: Optics Interface Error Information, Description: This command displays error information about the optics interface, including error counters and error rates.
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Resolution Steps

Step 1: Step: Replace the SFP/XFP Interface, Command: replace optics interface 1/1/1, Description: Replace the SFP/XFP interface with a new one to resolve the issue. Ensure that the new interface is properly seated and secured.

Step 2: Step: Realign the Fiber Optic Cables, Command: realign optics interface 1/1/1, Description: Realign the fiber optic cables to ensure proper connection and alignment. Verify that the cables are securely connected to the SFP/XFP interface.

Step 3: Step: Adjust the Rx/Tx Power Levels, Command: adjust optics interface 1/1/1 tx_power 10, Description: Adjust the receive/transmit power levels to ensure optimal signal transmission and reception. Verify that the power levels are within the acceptable range.

Post-Check Verification

```
Command: display optics interface brief, Expected Output: Optics Interface Brief Information, Description: Verify that the optics interface is functioning properly and that the link status is up.

Command: display optics interface detailed, Expected Output: Optics Interface Detailed Information, Description: Verify that the receive/transmit power levels are within the acceptable range and that there are no errors.
```

Roll-back Procedures

- Step: Restore the Original SFP/XFP Interface, Command: undo replace optics interface 1/1/1, Description: Restore the original SFP/XFP interface to its previous state in case the new interface does not resolve the issue.

- Step: Revert the Rx/Tx Power Levels, Command: undo adjust optics interface 1/1/1 tx_power 10, Description: Revert the receive/transmit power levels to their previous state in case the adjustment does not resolve the issue.